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FULL STACK APPLICATION DEVELOPMENT (24SDCS02)

2025-26, Even Semester

Project Problem Statement Submission Form

Sec No-05	**Team No-13**	**Project Title-Personal Knowledge Capsule**

1.Abstract

In today's digital environment, individuals continuously generate personal knowledge in the form of notes, ideas, study materials, links, and reflections. However, this information is often stored in an unorganized manner across notebooks, mobile applications, and browsers, making retrieval and management difficult. The lack of a centralized and structured system results in loss of valuable information and reduced productivity.

The **Personal Knowledge Capsule** is a full-stack web application designed to provide a centralized platform for storing, organizing, and retrieving personal knowledge efficiently. The system allows users to create knowledge entries and organize them using categories and moods, enabling context-based retrieval. Users can easily filter and sort their stored knowledge based on different parameters such as category, mood, and creation date.

The application is developed using **Spring Boot** for backend services and **Hibernate with JPA** for database interaction, ensuring efficient data persistence and management. RESTful APIs are used to perform CRUD operations, making the system scalable and easy to integrate with front-end applications. Maven is used for project management and dependency handling, while Git is used for version control.

The proposed system provides a simple, efficient, and user-friendly solution for personal knowledge management

2.Module Mapping

Module Name	Assigned Roll Number(s)	Signature
User & Category Management Module	2420030014	
Knowledge Entry Management Module	2420030368	
Search, Filter & API Integration Module	2420030573	

3.Module Details

Member 1 – User & Category Management Module
Responsible for designing and implementing user-related functionalities such as creating and retrieving user details. This member also handles the category management module, which includes creating, viewing, and managing categories used to organize knowledge entries. Database tables and JPA entity mappings for users and categories are handled in this module.

Member 2 – Knowledge Entry Management Module
Responsible for implementing the core functionality of the application. This includes adding, updating, deleting, and viewing knowledge entries. Each knowledge entry contains details such as title, content, mood, category, and creation date. This member ensures proper Hibernate and JPA mappings between knowledge entries, users, and categories.

Member 3 – Search, Filter & API Integration Module
Responsible for implementing RESTful APIs to retrieve knowledge entries based on filters such as mood, category, and date. This module uses Spring Data JPA derived query methods and JPQL queries. The member also ensures proper API responses using ResponseEntity and handles basic exception handling.

4.Frameworks / Platforms / Software Used

- **Programming Language:** Java
- **Backend Framework:** Spring Boot
- **ORM Framework:** Hibernate with JPA
- **Database:** MySQL / PostgreSQL
- **Build Tool:** Maven
- **Version Control:** Git
- **IDE:** Spring Tool Suite (STS) / Eclipse
- **API Testing Tool:** Postman

Guide Name and Signature (remarks if any) _____